

Cody Lee

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Building end-to-end LLM pipelines — retrieval, agent loops, evaluation, and cost-tiered routing — and forensically documenting where they break across trading systems, financial documents, and production stacks.

Education

University of California, Riverside • B.S. Computer Science

Expected Jun 2026

Relevant coursework: Natural Language Processing, Artificial Intelligence, Data Science, Probability & Statistics

Projects

Aether — LLM Workflow Engine

Apr 2026

- Built a **reason-act-observe agent** that turns a financial document and a plain-language goal into a **grounded chart** where every value traces to a source cell — direct SDK, no LangChain.
- **75.5% end-to-end on FinQA** (n=200); retrieval R@5 0.85, MRR@3 0.73 over 200 queries; six-tool loop with per-step Pydantic validation.
- Aggregated a **700-row financial dataset** into per-segment SQL totals matching an independent computation exactly; full **observability** — every step and tool call traced to SQLite and rendered as a visual reasoning trace.

Production RAG Stack Forensics [5-page autopsy](#)

2026

- Built a **production RAG system** on the industry-standard stack — LangGraph, Pinecone, Langfuse, and cross-provider inference (Claude, GPT-5.5, Gemini); documented **four failure modes** with measured fixes against a 150-question eval suite.
- Caught an **evaluation bug** scoring answers against empty retrieved content — hidden until a different model exposed it — that had inverted a result; the fix reversed it, preventing a wrong published finding.
- Built a custom **MCP server** exposing eval, latency, cost, and failure-mode data as agent-queryable tools over the open Model Context Protocol.

Polymarket Trading Bot Autopsy [2-page summary](#) | [15-page autopsy](#)

Apr–Jun 2026

- Found **three measurement bugs** that inflated paper trading results $\sim 135\times$ and made the strategy misleading — the bots flagged as best performed worst live.
- Built a **3-layer LLM classification pipeline** (Haiku → Sonnet → Opus) over 2,409 wallets; 180 bots, 417K simulated trades. Tiered routing cut cost to \$30–40 vs $\sim \$150$ on Opus alone.
- Operated a **live execution environment** with wallet auth, kill switches, and trade logging; reconciled millions of cross-source transactions across Polymarket, Binance, and on-chain data.

ChainTax — Crypto Tax Engine

Feb–Mar 2026

- Built a **cross-source ingestion and reconciliation pipeline**: 540K+ events across Hyperliquid and two on-chain APIs (Alchemy, DeBank), with bridge detection to prevent double-counting across chains.
- **FIFO lot matching** with three competing IRS funding treatments producing three Form 8949 variants per run, plus offsetting-position detection across spot, perp, and correlated pairs.

Technical Skills

Languages Python, TypeScript, SQL, Bash

LLM / RAG Hybrid retrieval (dense + sparse), chunking, embedding, reranking, vector databases (Pinecone, ChromaDB), prompt engineering, LLM evaluation, agent orchestration (LangGraph), MCP servers, Anthropic / OpenAI / Google SDKs

Observability Langfuse tracing, per-stage cost / latency instrumentation

Data Cross-source ingestion and reconciliation, pandas, DuckDB, SQLite, async websocket pipelines

Infrastructure Git, REST APIs, FastAPI, Streamlit, VPS deployment

Methodology Forensic measurement, failure-mode analysis, statistical evaluation, technical documentation